Mastering MLP

Computation Intelligence and its Application in Mechatronics

Winter 2025

Amirkabir University of Technology
(Tehran Polytechnic)

The Vanishing/Exploding Gradient Problem

- **Issue:** As gradients backpropagate, they may shrink to near-zero (vanishing) or grow too large (exploding).

- Effect:
  - Vanishing gradients slow down learning or prevent deeper layers from learning.
  - Exploding gradients cause instability and divergence.

The Vanishing/Exploding Gradient Problem

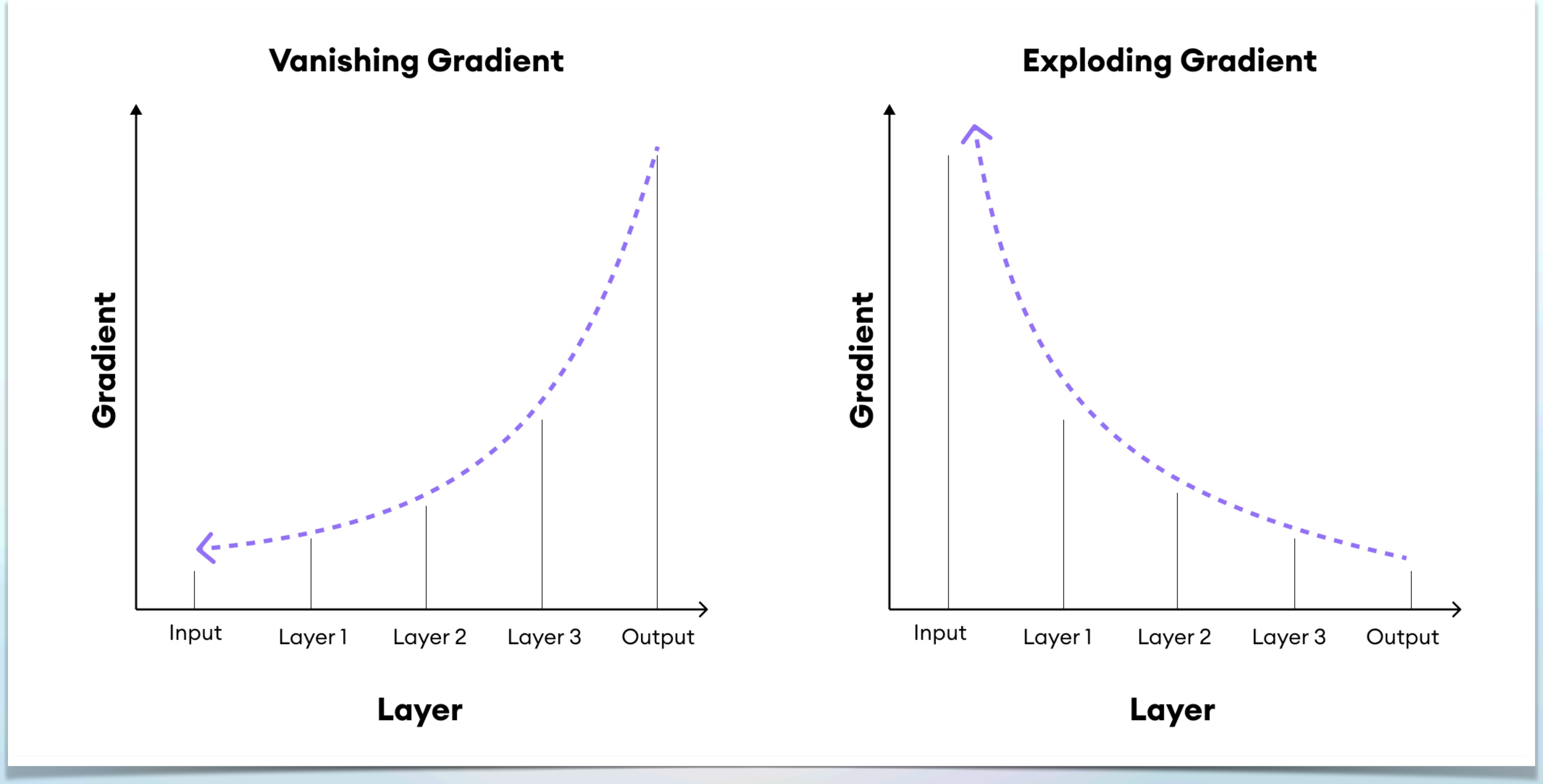


Image From: https://www.superannotate.com/blog/activation-functions-in-neural-networks

The Vanishing/Exploding Gradient Problem

Solutions to Vanishing/Exploding Gradients

1. Better Weight Initialization (Glorot, He)
2. Nonsaturating Activation Functions (Leaky ReLU, etc.)
3. Batch Normalization
4. Gradient Clipping

The Vanishing/Exploding Gradient Problem

Weight Initialization Techniques

Glorot (Xavier) Initialization

- Formula: Uniform Distribution $w \sim U\left(-\frac{\sqrt{6}}{\sqrt{n_{\text{in}} + n_{\text{out}}}}, \frac{\sqrt{6}}{\sqrt{n_{\text{in}} + n_{\text{out}}}}\right)$, Normal Distribution $w \sim \mathcal{N}\left(0, \frac{2}{n_{\text{in}} + n_{\text{out}}}\right)$
- Suitable for **sigmoid** and **tanh** activations.

He Initialization

- Formula: Uniform Distribution $w \sim U\left(-\frac{\sqrt{6}}{\sqrt{n_{\text{in}}}}, \frac{\sqrt{6}}{\sqrt{n_{\text{in}}}}\right)$, Normal Distribution $w \sim \mathcal{N}\left(0, \frac{2}{n_{\text{in}}}\right)$
- Best for **ReLU** and **Leaky ReLU** activations

The Vanishing/Exploding Gradient Problem

Nonsaturating Activation Functions

Leaky ReLU (LReLU) $= \begin{cases} x, & \text{if } x > 0 \\ \alpha x, & \text{if } x \leq 0 \end{cases}$

Randomized Leaky ReLU (RReLU) (randomizes α during training)

Parametric Leaky ReLU (PReLU) (learns the negative slope α during training)

Exponential Linear Unit (ELU) $= \begin{cases} x, & x > 0 \\ \alpha(\exp(x) - 1), & x \leq 0 \end{cases}$

Scaled Exponential Linear Unit (SELU)

The Vanishing/Exploding Gradient Problem

Nonsaturating Activation Functions

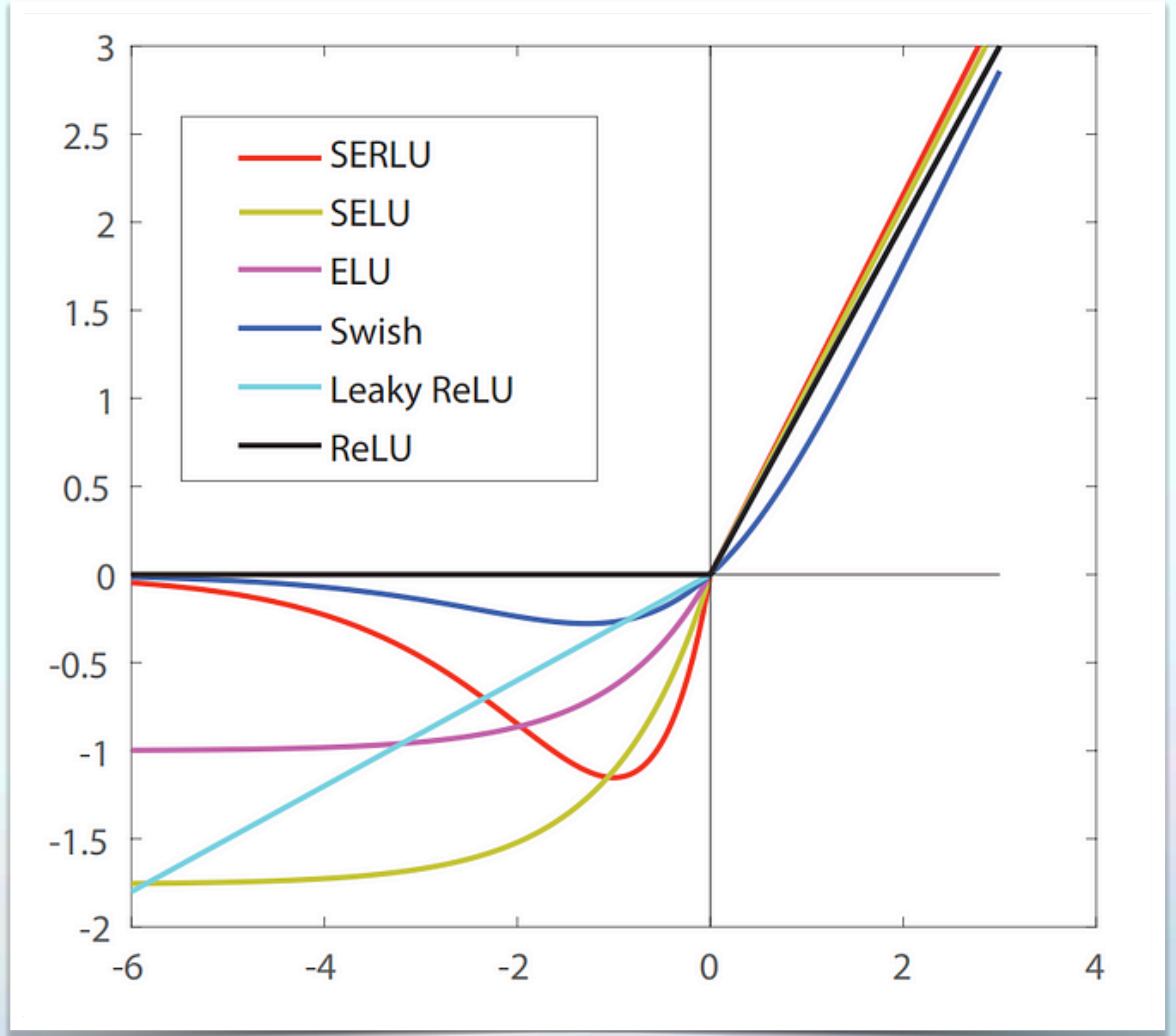


Image From: https://medium.com/towards-data-science/a-comprehensive-guide-on-activation-functions-b45ed37a4fa5

The Vanishing/Exploding Gradient Problem

Batch Normalization

Normalizes activations across the mini-batch:

$$\hat{x}_i = \frac{x_i - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}}, \qquad z_i = \gamma \hat{x}_i + \beta$$

$$\mu_B = \frac{1}{m}\sum_{i=1}^{m} x_i, \qquad \sigma_B^2 = \frac{1}{m}\sum_{i=1}^{m} (x_i - \mu_B)^2$$

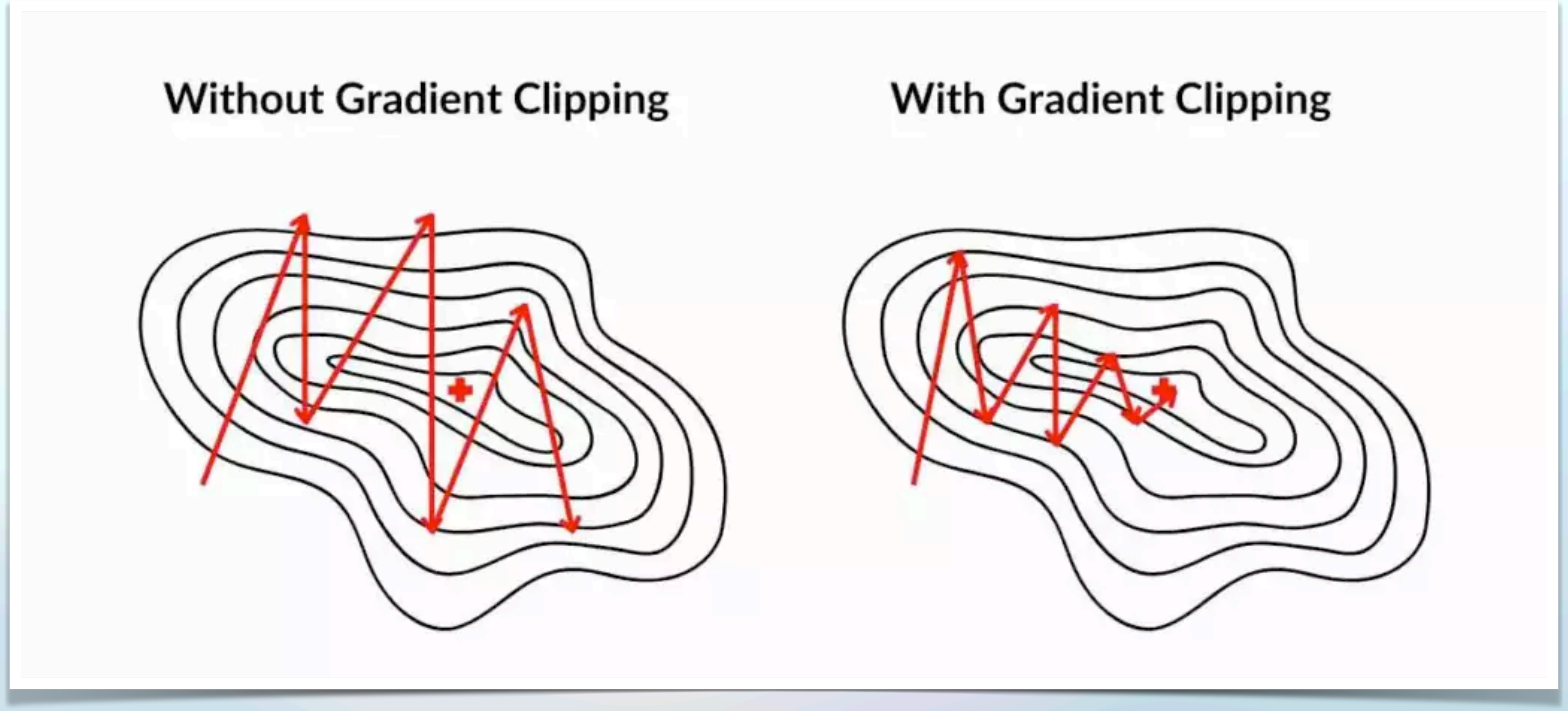
- x_i is the input activation of the neuron.
- μ_B is the mean of the mini-batch.
- σ_B^2 is the variance of the mini-batch.
- $\hat{x}_i$ is the normalized activation.
- ϵ is a small constant to prevent division by zero.
- γ (scale) and β (shift) are trainable parameters.

Image From: https://www.vecteezy.com/

The Vanishing/Exploding Gradient Problem

Gradient Clipping

Prevents exploding gradients by limiting their magnitude.



mage From: https://spotintelligence.com/2023/12/01/gradient-clipping-explained-practical-how-to-guide-in-python/

Avoiding Overfitting – Regularization Techniques

L1 and L2 regularization

L1 regularization: lasso regression

$$\text{New Cost} = \frac{1}{n}\sum_{n}^{i} y_i - \hat{y}_i + \lambda \sum_{i} |w_i|$$

L2 regularization: ridge regression

$$\text{New Cost} = \frac{1}{n}\sum_{n}^{i} y_i - \hat{y}_i + \lambda \sum_{i} w_i^2$$

Avoiding Overfitting – Regularization Techniques

Dropout

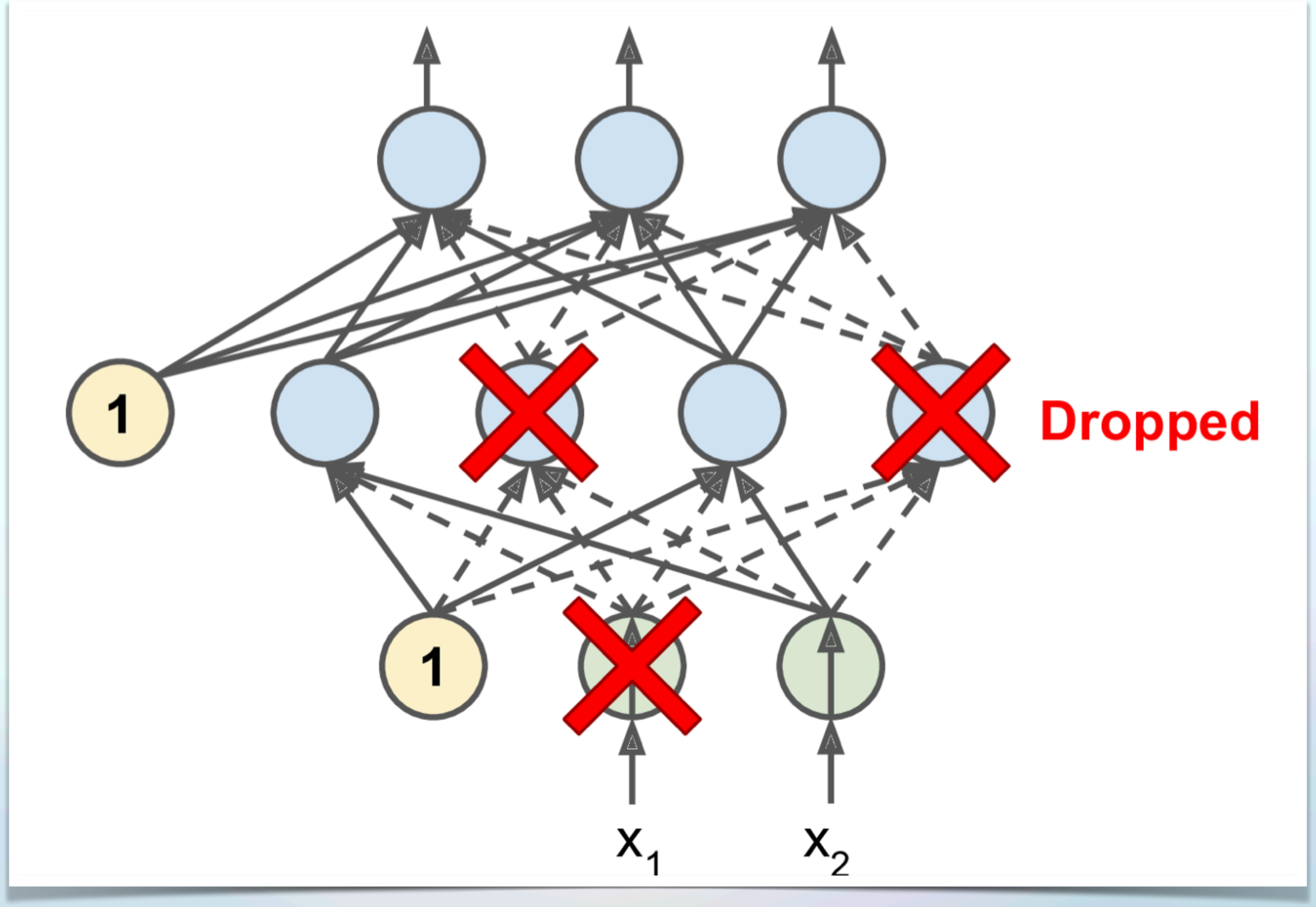


Image from: Hands-On Machine Learning with Scikit-Learn, Keras, and Tensorflow

Optimizing Learning Process

Faster Optimizers

Momentum Optimization

- Adds velocity to updates

$$v_t = \beta v_{t-1} + (1 - \beta)\nabla J(\theta)$$

$$\theta_t = \theta_{t-1} - \eta v_t$$

where β is the momentum coefficient

Stochastic Gradient Descent (SGD)

- Stochastic Gradient Descent (SGD) is the most basic optimization algorithm, updating parameters using the gradient of the loss function:

$$\theta_t = \theta_{t-1} - \eta \nabla J(\theta_{t-1})$$

Vanilla SGD is not enough!

We need better optimization strategies.

Faster Optimizers

Nesterov Accelerated Gradient (NAG)

- NAG improves momentum by computing the gradient at the estimated future position

$$v_t = \beta v_{t-1} + (1 - \beta) \nabla J(\theta_{t-1} - \eta \beta v_{t-1})$$

$$\theta_t = \theta_{t-1} - \eta v_t$$

Faster Optimizers

RMSProp (Root Mean Square Propagation)

- RMSProp adapts the learning rate for each parameter based on the moving average of squared gradients

$$s_t = \beta s_{t-1} + (1 - \beta)(\nabla J(\theta))^2$$

$$\theta_t = \theta_{t-1} - \frac{\eta}{\sqrt{s_t + \epsilon}} \nabla J(\theta)$$

Where β is the decay factor

And ϵ is a small constant for numerical stability.

Faster Optimizers

Adam (Adaptive Moment Estimation)

- Adam combines **Momentum** and **RMSProp:**

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1)\nabla J(\theta)\,, \quad s_t = \beta s_{t-1} + (1 - \beta)(\nabla J(\theta))^2$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}\,, \quad \hat{s}_t = \frac{s_t}{1 - \beta_2^t}$$

$$\theta_t = \theta_{t-1} - \frac{\eta}{\sqrt{\hat{s}_t} + \epsilon}\hat{m}_t$$

Faster Optimizers

Nadam (Nesterov-Accelerated Adaptive Moment Estimation)

- Nadam combines **Adam** with Nesterov **momentum**:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) \nabla J(\theta)\,, \quad s_t = \beta s_{t-1} + (1 - \beta)(\nabla J(\theta))^2$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}\,, \quad \hat{s}_t = \frac{s_t}{1 - \beta_2^t}$$

$$\theta_t = \theta_{t-1} - \frac{\eta}{\sqrt{\hat{s}_t} + \epsilon}(\beta_1 \hat{m}_t + (1 - \beta_1) \nabla J(\theta))$$

Learning Rate Scheduling

- Time-based decay : $\eta_t = \dfrac{\eta_0}{1 + kt}$
- Step decay: $\eta_t = \eta_0 \cdot \gamma^{(t/s)}$
- Exponential decay: $\eta_t = \eta_0 e^{-kt}$

Next, we will explore even more advanced architectures and techniques!